

K-FIELD

LUCAS SUPERGROUP EXTRACTED DEFINITION

A stand-alone foundational and archival reference defining the 9B Lucas Supergroup from the authoritative Cell3 basis: axioms, metric geometry, complete node and plane structure, balanced-ternary aggregation, cellular topology, laminar laws, shell arithmetic, reciprocal spectra, principal modes, and exact closure identities.

Exact deterministic hierarchy generated from the authoritative Cell3 basis

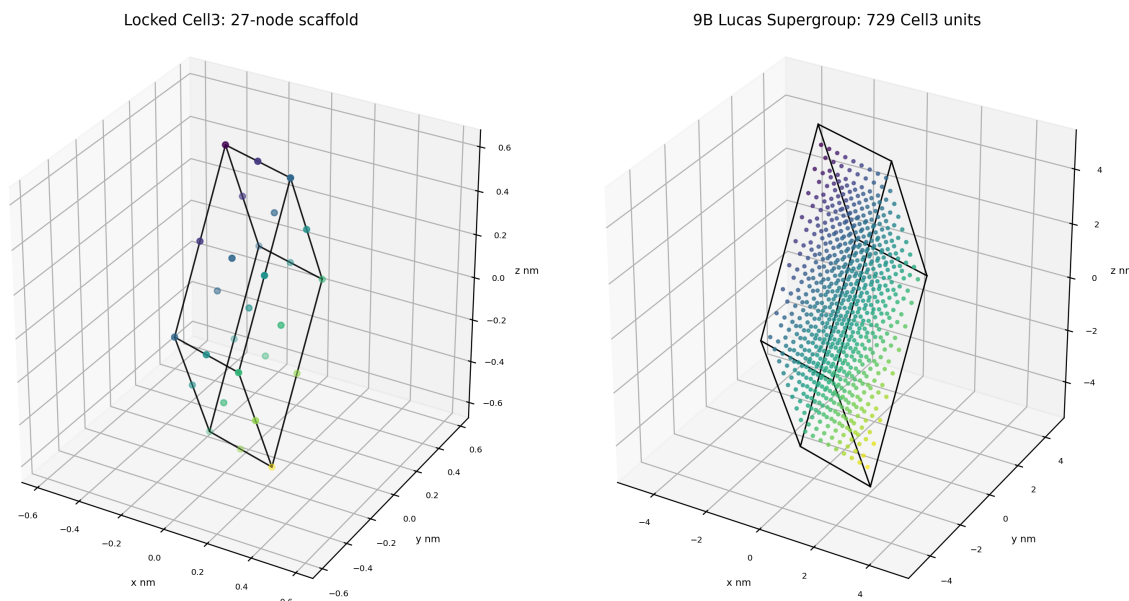


Figure 1. Title composite generated directly from the locked Cell3 basis. The left panel is the 27-node primitive scaffold. The right panel is the dimensionally correct 9B block containing 729 Cell3 centers.

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Status and interpretation contract

This paper defines a finite mathematical object and its exact numerical realization. Statements derived solely from the locked basis, integer address sets, matrix identities, incidence counts, or explicitly reconstructed graph operators are presented as exact geometric or algebraic results. The magnetic-moment comparison is retained as a numerical candidate identification because the source does not contain a dynamical coupling law that derives that observable. The distinction between exact structure and candidate physical interpretation is part of the definition, not an editorial qualification.

Central result

The Lucas Supergroup is the unique positive $m \times m \times m$ Cell3 block for which the complete half-step boundary-state count equals the internal Cell3 interface count plus the Euler characteristic of its spherical boundary: $1946 = 1944 + 2$. At $m=9$ it contains 729 Cell3 volumes, 27 Lucas Groups, 1000 unrefined vertices, 6859 complete barycentric states, 55 refined laminar sheets, 3429 antipodal refined pairs, and 864 primitive antipodal outer-shell directions.

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Abstract

The Lucas Supergroup is a finite, metric-weighted, balanced-ternary cubical complex generated from one authoritative oblique Cell3 basis. A Cell3 supplies a 27-node half-step scaffold, a direct metric, a reciprocal metric, six boundary planes, a laminar grade, a parity character, and an antipodal involution. A first ternary aggregation forms a Lucas Group containing 27 complete Cell3 units. A second ternary aggregation forms the 9B Lucas Supergroup containing 27 Lucas Groups and 729 Cell3 volumes. The resulting object is simultaneously a physical parallelepiped in ICRS coordinates, a cubical cellular complex, a refined $19 \times 19 \times 19$ barycentric state lattice, a modulo-nine phase system, a family of constant-sum laminar sheets, a recursive shell hierarchy, and a finite reciprocal directional spectrometer.

The complete complex has f-vector (1000,2700,2430,729) and Euler characteristic 1. Its strict interior has f-vector (512,1728,1944,729), while its boundary has f-vector (488,972,486) and Euler characteristic 2. The equality $1946=1944+2$ independently selects $m=9$. The 1944 internal Cell3 interfaces split exactly as 1458 intragroup, 486 intergroup, and 486 external faces, giving the ratio 3:1:1. The 729-cell adjacency graph has cycle rank 1216. The refined lattice contains 55 sheets with exact population law $P_9(\lambda)=271-\lambda^2$ in the central region and triangular tails outside it. The primitive radius-nine shell contains 864 antipodally distinct directions, and its reciprocal q-observations span the full six-dimensional space of symmetric three-dimensional metrics.

This paper reconstructs the complete canonical definition, derives every central formula, explains the meaning of every census, reproduces all 27 primitive nodes and six boundary planes, and records the limits of what is and is not fixed by the source. The result is a self-contained reference suitable for technical instruction, computational reproduction, and further K-Field development.

1. Reader Orientation and Notation

The term **Cell3** denotes the primitive oblique three-dimensional cell and its complete 27-node half-step scaffold. The term **Lucas Group** denotes a $3 \times 3 \times 3$ assembly of Cell3 volumes. The term **Lucas Supergroup** denotes a $9 \times 9 \times 9$ assembly of Cell3 volumes, equivalently a $3 \times 3 \times 3$ assembly of Lucas Groups. The symbol B denotes the ordered direct basis; $M=B^T B$ is its direct metric; $R=M^{-1}$ is the reciprocal metric. Integer triples are called addresses. Their component sum is the laminar index λ , and the parity character $\chi=(-1)^\lambda$ assigns binary polarity.

Symbol	Meaning
$B=[a \ b \ c]$	Authoritative ordered Cell3 basis matrix; columns are direct basis vectors
$M=B^T B$	Direct metric in nm^2
$R=M^{-1}$	Reciprocal metric in nm^{-2}
$u=(u_1, u_2, u_3)$	Integer half-step or reciprocal address
$r(u)=B \ u / 2$	Physical position of a half-step address
$x=3G+c$	Balanced-ternary Cell3-center address in the Supergroup
$\lambda(u)=u_1+u_2+u_3$	Laminar or constant-sum grade
$\chi(u)=(-1)^\lambda$	Binary polarity character
$A:u \rightarrow -u$	Antipodal involution
$F_m(t)=((m+1)+mt)^3$	Generating polynomial of the $m \times m \times m$ cubical complex
S_r	Integer Chebyshev shell $\ u\ _\infty=r$
$q(u)=u^T R \ u$	Reciprocal quadratic value

1.1 Levels of description

The same finite object is used in several compatible descriptions. In direct geometry it is a parallelepiped with measured lengths, face areas, volume, and face spacings. In cellular topology it is a finite cubical complex with vertices, edges, faces, and volumes. In the half-step description it is a $19 \times 19 \times 19$ lattice whose points represent barycenters of all cellular dimensions. In the phase description its periodic address classes form Z_9^3 . In reciprocal geometry every integer direction receives a metric value $q(u)$. These descriptions must not be conflated: for example, the 6859 half-step states are not 6859 ordinary vertices, and the 55 refined laminar sheets are not the same as the 25 sheets occupied by the 729 Cell3 centers.

2. Primitive Cell3 Geometry

2.1 Authoritative direct basis

The Cell3 geometry begins with the ordered basis matrix B. Its three columns are the primitive vectors a, b, and c. All physical coordinates in this paper are generated from this matrix; no illustrative deformation or orthogonal replacement is used.

Component	a (nm)	b (nm)	c (nm)
ICRS x	0.327501242055891	0.035330775843828	-0.010129317626729
ICRS y	-0.054897007884006	0.207319850440192	-0.389787702366260
ICRS z	-0.037487502894730	-0.547401085388508	-0.689734408367268

$$B = \begin{bmatrix} 0.3275012420558906, & 0.0353307758438284, & -0.01012931762672882 \\ -0.0548970078840064, & 0.2073198504401918, & -0.38978770236625987 \\ -0.0374875028947299, & -0.5474010853885081, & -0.6897344083672676 \end{bmatrix} \text{ nm}$$

2.2 Direct and reciprocal metrics

The direct metric records all dot products among the basis vectors. It is positive definite and therefore determines lengths, angles, areas, volume, and the reciprocal metric without additional geometric input.

Direct metric M in nm²

	1	2	3
1	0.111676057896050	0.020710333280606	0.043937235096220
2	0.020710333280606	0.343877732392787	0.296392758961523
3	0.043937235096220	0.296392758961523	0.627770610077296

Reciprocal metric R in nm⁻²

	1	2	3
1	9.208025529457549	0.001534802428193	-0.645188035699939
2	0.001534802428193	4.903398312367987	-2.315175584416389
3	-0.645188035699939	-2.315175584416389	2.731171274711695

$$M = B^T B, \quad R = M^{-1}, \quad M R = I.$$

The reciprocal metric is not a second independent fit. It is the exact inverse of the direct metric. Every reciprocal-shell result later in the paper is therefore tied to the same primitive geometry that fixes the Cell3 dimensions.

2.3 Lengths, angles, areas, volume, and spacings

Quantity	Value	Definition
a	0.3341796790591105 nm	$\sqrt{M11}$
b	0.5864108904111408 nm	$\sqrt{M22}$
c	0.7923197650426852 nm	$\sqrt{M33}$
theta_ab	83.93348301231273 deg	$\arccos(M12/(a b))$
theta_ac	80.44813001833790 deg	$\arccos(M13/(a c))$
theta_bc	50.36323181375618 deg	$\arccos(M23/(b c))$
A_ab	0.1948691654604841 nm ²	a x b
A_ac	0.2611062357902053 nm ²	a x c
A_bc	0.3578095391285615 nm ²	b x c
V1	0.1179148919117447 nm ³	det B
h_a	0.3295465296954468 nm	$V1/A_{bc}$
h_b	0.4515973797212847 nm	$V1/A_{ac}$
h_c	0.6050977415185554 nm	$V1/A_{ab}$
d1	1.3436536216759252 nm	a+b+c

The Cell3 is strongly oblique. The quoted body diagonal is the diagonal generated by a+b+c and is the longest of the four inequivalent absolute diagonal lengths. This distinction matters because an oblique parallelepiped does not have one universal body-diagonal length under all sign combinations.

Locked Cell3 geometry and node-address index

Dimensionally correct Cell3: all 27 nodes

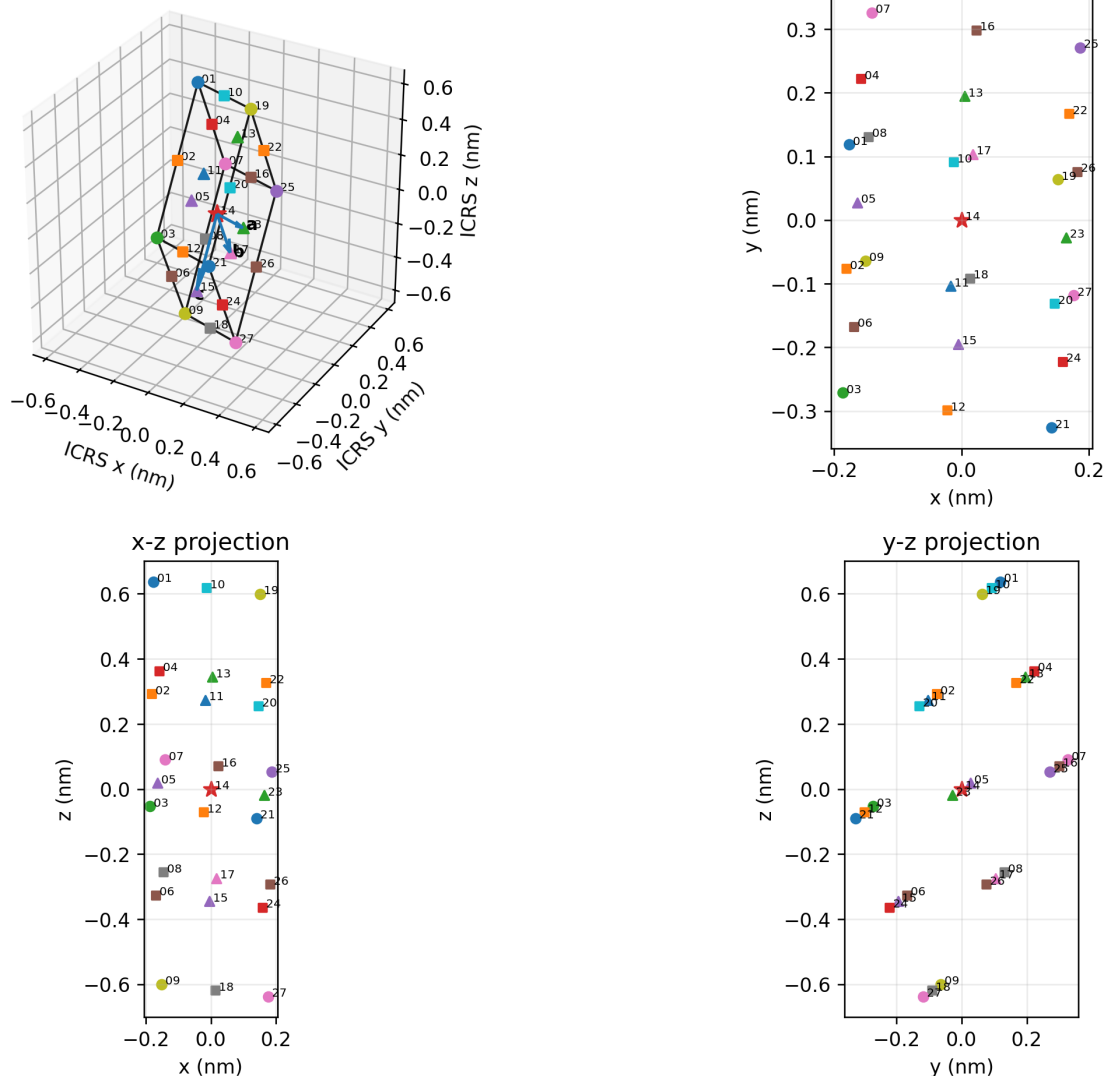


Figure 2. Complete dimensionally correct Cell3. Node labels 01-27 correspond to the full node ledger in Appendix A. The three orthographic projections preserve the ICRS dimensions and make coincident depth relationships visible.

3. The 27-Node Cell3 Scaffold

3.1 Half-step embedding axiom

Every local scaffold node is identified by an address $u=(u1,u2,u3)$ with each coordinate in $\{-1,0,1\}$. The physical embedding is

$$r(u) = (1/2) B u, \quad u \in \{-1,0,1\}^3.$$

The address cube has $3^3=27$ elements. The factor one-half places the nonzero coordinate values on opposite primitive faces and places zeros at the corresponding midplanes. The construction therefore includes corners, edge centers, face centers, and the body center in one uniform formula.

3.2 Node class from zero count

Number of zero coordinates	Node class	Population
0	Corner	8
1	Edge center	12
2	Face center	6
3	Body center	1

The class count is the coefficient structure of $(2+1)^3$. Choosing which coordinates are zero and assigning signs to the remaining coordinates gives $8+12+6+1=27$.

3.3 Laminar grade, polarity, and antipodal symmetry

$$\lambda(u) = u_1 + u_2 + u_3, \chi(u) = (-1)^{\lambda(u)}, A(u) = -u.$$

For the primitive scaffold, λ ranges from -3 to $+3$, giving seven constant-sum sheets. Antipodal inversion sends λ to $-\lambda$ while leaving parity unchanged, because $(-1)^{-\lambda} = (-1)^{\lambda}$. Therefore antipodal partners occupy opposite laminar sheets but carry the same binary polarity. This pairing is exact for every node except the body center, which is fixed by inversion.

The complete numerical node ledger appears in Appendix A. It records the address, laminar index, polarity, geometric class, and ICRS position of every node. The figure and ledger together satisfy two different instructional needs: the figure shows the oblique spatial arrangement, while the ledger supplies exact reproducible coordinates.

4. Six Boundary Planes

4.1 Dual-coordinate plane equations

The direct basis defines a dual coordinate system through B^{-1} . The two faces normal to each primitive channel are the level sets

$$e_i^T B^{-1} r = \pm 1/2, i \in \{a, b, c\}.$$

The a-channel planes bound the bc faces, the b-channel planes bound the ac faces, and the c-channel planes bound the ab faces. Every plane contains exactly nine of the 27 scaffold nodes because one address coordinate is fixed and the other two independently take three values.

Channel	Face family	Unit normal in ICRS	Face area (nm ²)	Opposite spacing (nm)	Center-face distance (nm)
a	bc	(-0.995965190, 0.083602163, -0.032619291)	0.357809539128562	0.329546529695447	0.164773264847723
b	ac	(-0.089052594, -0.866576769, 0.491033948)	0.261106235790205	0.451597379721285	0.225798689860642
c	ab	(0.194092201, 0.913177169, 0.358379235)	0.194869165460484	0.605097741518555	0.302548870759278

A plane normal and its negative represent the same unoriented plane. The stored normal signs are therefore a convention. For outward arrows on opposite faces, the normal is multiplied by the face address coefficient. The center-to-face distances are exactly one-half of the opposite-face spacings.

Six Cell3 boundary planes: exact nine-node incidence and oriented normals

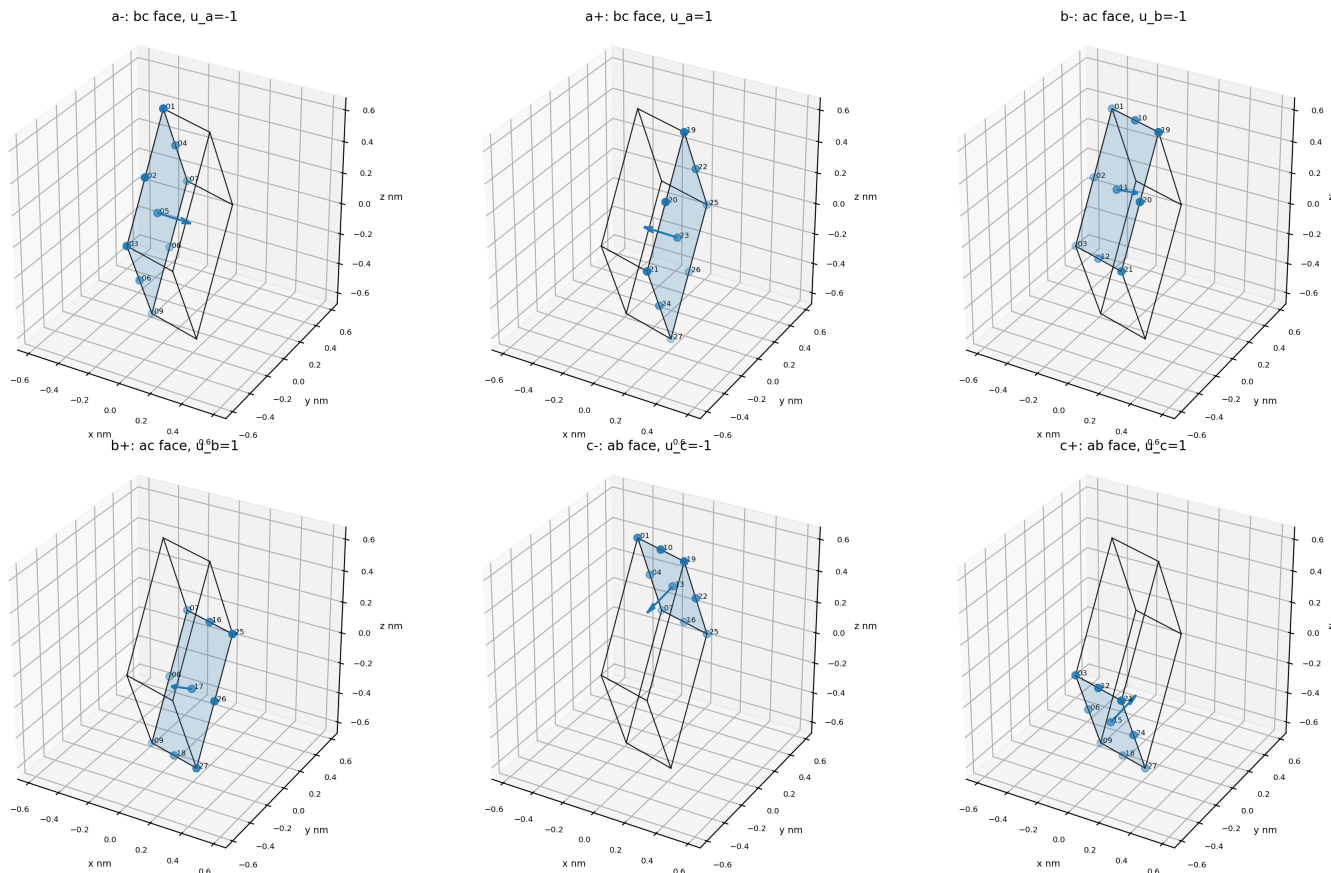


Figure 3. Six boundary planes. Each panel highlights the exact nine-node incidence set, the physical face polygon, and the outward-oriented normal used for that face. Node numbers reference Appendix A.

5. Lucas Group and Lucas Supergroup Composition

5.1 Balanced-ternary address axiom

The Supergroup center address x is decomposed uniquely into a group address G and an intra-group Cell3 address c :

$$x = 3G + c, \quad G \in \{-1, 0, 1\}^3, \quad c \in \{-1, 0, 1\}^3.$$

The vector G chooses one of 27 Lucas Groups. The vector c chooses one of 27 Cell3 units inside that Lucas Group. Because each coordinate has a unique balanced-ternary representation $3G_i + c_i \in \{-4, -3, \dots, 4\}$, the complete center-address set is $X_9 = \{-4, -3, \dots, 4\}^3$ and contains $9^3 = 729$ elements.

$$|X_9| = 9^3 = 729 = 27 \text{ Lucas Groups} \times 27 \text{ Cell3 units per Lucas Group}.$$

5.2 Physical embedding of centers

$$r_{\text{center}}(x) = B \cdot x.$$

The center-address formula does not contain the one-half factor used for barycentric scaffold nodes. Neighboring Cell3 centers differ by one full basis vector. Each volume then extends one-half basis step in each primitive direction. Therefore the outer boundary of the 9B block occurs at primitive coordinates ± 4.5 .

5.3 Laminar and polarity factorization

$$\lambda_x = 3 \lambda_G + \lambda_c, \quad \chi(x) = \chi(G) \chi(c).$$

The laminar decomposition follows by summing the components of $x=3G+c$. The polarity factorization follows because three is odd, so $(-1)^{(3\lambda_G)}=(-1)^{\lambda_G}$. The recursive ternary address therefore preserves a simple multiplicative parity character across levels.

5.4 Periodic phase group

When addresses are interpreted periodically modulo nine, the phase classes form Z_9^3 . The associated exact sequence is

$$0 \rightarrow Z_3^3 \rightarrow Z_9^3 \rightarrow Z_3^3 \rightarrow 0.$$

The extension does not split. Z_9 contains elements of order nine, whereas a direct product $Z_3 \times Z_3$ has exponent three. This group statement applies to modulo-nine phase addresses. The bounded set $\{-4, \dots, 4\}^3$ is a representative cube and is not closed under ordinary integer addition.

Lucas Supergroup: $9 \times 9 \times 9$ Cell3 block with $3 \times 3 \times 3$ Lucas-Group partition

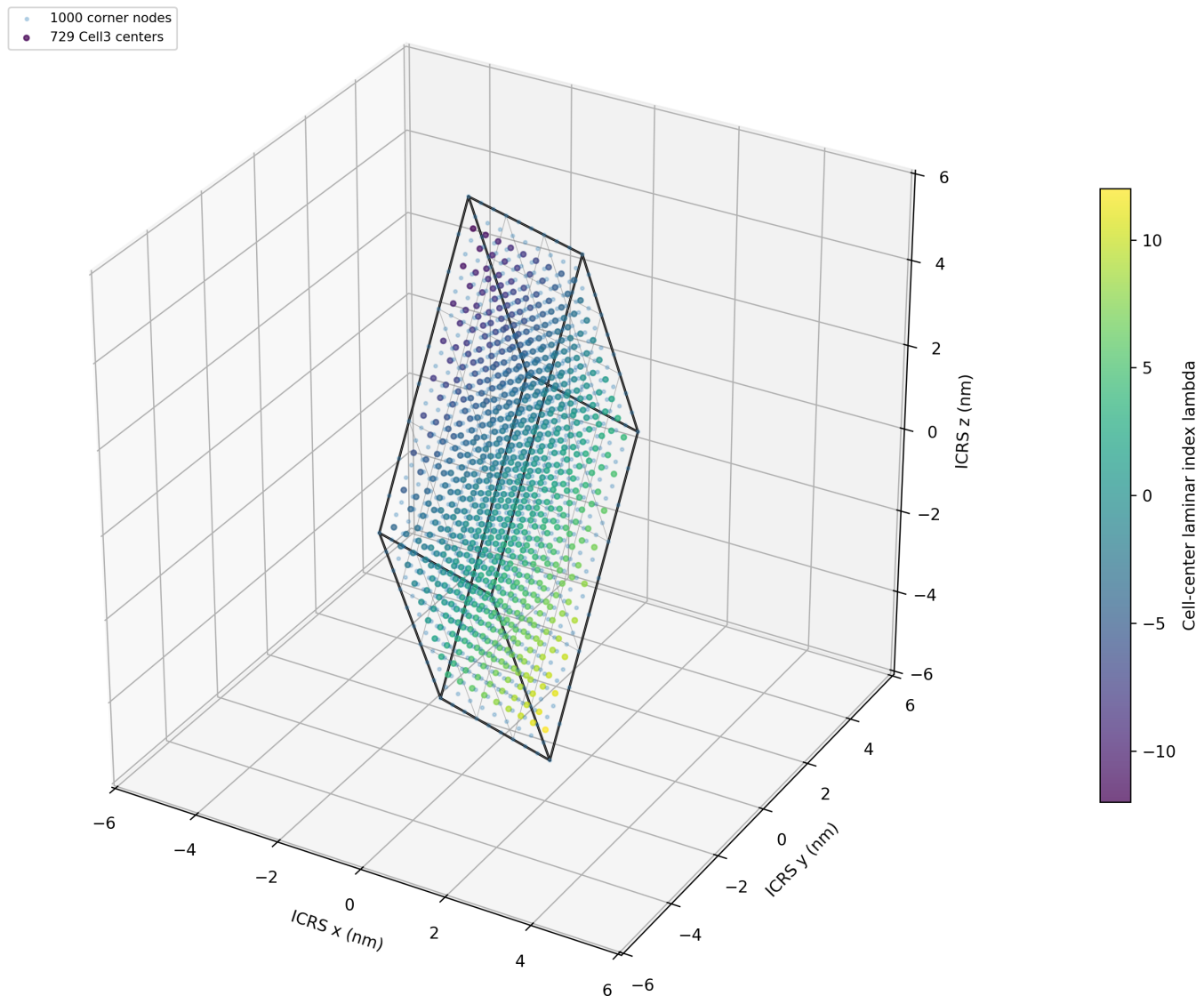


Figure 4. Dimensionally correct Lucas Supergroup. The 729 Cell3 centers and 1000 merged corner nodes are generated from B. The heavy outline is the 9B boundary; the surface grid marks the 3B Lucas-Group partition. Center color is the exact laminar index.

6. Scale Hierarchy and Physical Dimensions

6.1 Similarity laws

For any linear scale factor s applied to the primitive basis, direct lengths scale as s , areas as s^2 , volumes as s^3 , and reciprocal metric values as s^{-2} :

$$B_s = sB, \quad M_s = s^2 M, \quad R_s = s^{-2} R, \quad A_s = s^2 A_1, \quad V_s = s^3 V_1.$$

6.2 Lucas Group at 3B

Quantity	3B value
Cell3 volumes	27
Edge lengths	1.002539037177; 1.759232671233; 2.376959295128 nm
Face areas	1.753822489144; 2.349956122112; 3.220285852157 nm ²
Volume	3.183702081617 nm ³
Reported body diagonal	4.030960865028 nm

6.3 Lucas Supergroup at 9B

Quantity	Exact value
Lucas Groups	27
Cell3 volumes	729
Outer edge lengths	3.007617111531995; 5.277698013700268; 7.130877885384167 nm
Outer face areas	$A_{ab}=15.784402402299214$; $A_{ac}=21.149605099006632$; $A_{bc}=28.982572669413482$ nm ²
Volume	85.959956203661847 nm ³
Opposite-face spans	2.965918767259021; 4.064376417491562; 5.445879673666999 nm
Reported body diagonal	12.092882595083326 nm

Every Supergroup dimension in this table is an exact 9, 81, or 729 scaling of the corresponding Cell3 quantity. The physical block remains oblique; it is not replaced by an axis-aligned rectangular box.

7. Complete Cubical Cellular Complex

7.1 General generating polynomial

An $m \times m \times m$ assembly of cubical volumes has $m+1$ vertices per coordinate direction and m one-dimensional intervals per coordinate direction. The complete cell census is generated by

$$F_m(t) = ((m+1) + m t)^3.$$

The coefficient of t^k is the number of k -dimensional cells. Choosing t means choosing an interval in that coordinate; choosing the constant term means choosing a vertex coordinate. Expansion gives

$$f_0=(m+1)^3, \quad f_1=3m(m+1)^2, \quad f_2=3m^2(m+1), \quad f_3=m^3.$$

7.2 Supergroup f-vector

$$F_9(t) = (10+9t)^3 = 1000 + 2700t + 2430t^2 + 729t^3.$$

Dimension	Object	Count
0	Vertices	1000
1	Edges	2700
2	Faces	2430
3	Volumes	729

$$\chi = f_0 - f_1 + f_2 - f_3 = 1000 - 2700 + 2430 - 729 = 1.$$

Euler characteristic one is the expected value for a contractible finite three-dimensional block. This is the topological statement that the complete Supergroup has the homotopy type of a point even though its cell adjacency graph contains many cycles.

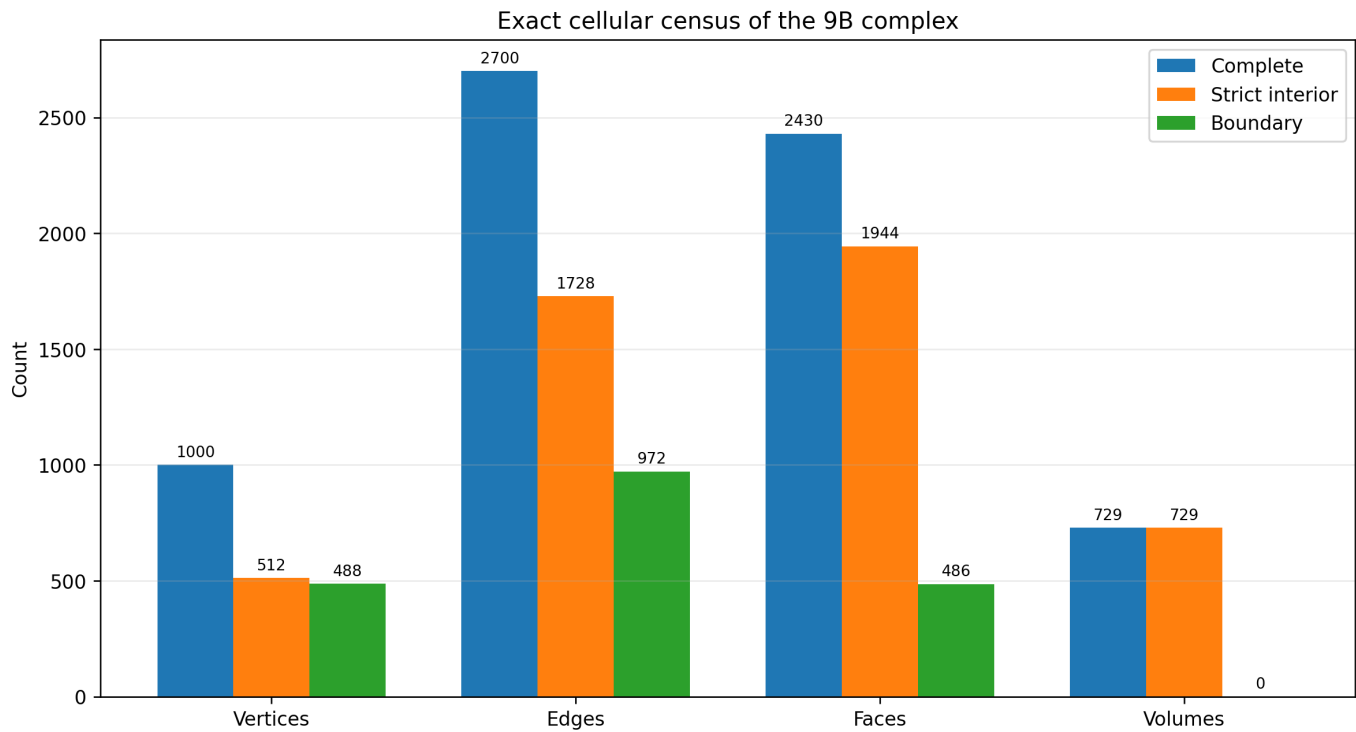


Figure 5. Exact complete, strict-interior, and boundary cell populations by dimension.

8. The Half-Step Lattice as Complete Cellular Census

8.1 Barycentric correspondence

Along one coordinate direction, the $m+1$ vertices and m interval barycenters interleave to form $2m+1$ half-step positions. Taking the Cartesian product across three coordinates gives $(2m+1)^3$ representative points. Each point represents exactly one cell of some dimension: zero odd half-step coordinates represent a vertex, one represents an edge barycenter, two represent a face barycenter, and three represent a volume center.

$$F_m(1) = ((m+1)+m)^3 = (2m+1)^3.$$

At $m=9$ the complete cellular-state lattice has 19 positions per axis and $19^3=6859$ states. The number 6859 is therefore the sum of all vertices, edges, faces, and volumes, not a count of ordinary vertices.

$$1000 + 2700 + 2430 + 729 = 6859 = 19^3.$$

8.2 Interior and boundary decomposition

Removing the outermost half-step layer leaves 17 positions per axis. The strict-interior generating polynomial is $(8+9t)^3$, because there are eight interior vertices and nine volume intervals per coordinate direction.

$$(8+9t)^3 = 512 + 1728t + 1944t^2 + 729t^3.$$

Complex	Vertices	Edges	Faces	Volumes	Total	Euler value
Complete	1000	2700	2430	729	6859	1
Strict interior	512	1728	1944	729	4913	-1 compact-support
Boundary	488	972	486	0	1946	2

The boundary counts are obtained by subtraction. Its Euler characteristic $488-972+486=2$ identifies a topological two-sphere. The compact-support Euler value -1 of the open interior is the three-dimensional sign expected for an open 3-cell.

9. Independent Interface-Boundary Selection of 9B

9.1 Boundary-state formula

The half-step boundary consists of all points in the $(2m+1)^3$ cube that are not in its $(2m-1)^3$ strict interior. Therefore

$$B_m = (2m+1)^3 - (2m-1)^3 = 24m^2 + 2.$$

9.2 Internal-interface formula

In each of the three coordinate directions, there are $m-1$ internal interface planes. Each contains m^2 Cell3 faces. Thus

$$I_m = 3m^2(m-1).$$

9.3 Unique positive closure solution

Demanding that the non-topological part of the boundary-state census equal the internal-interface count gives

$$B_m - 2 = I_m \quad 24m^2 = 3m^2(m-1) \quad 8 = m-1 \quad m = 9.$$

Because m is positive, division by $3m^2$ is valid. The solution is unique. At this scale,

$$B_9 = 1946, \quad I_9 = 1944, \quad B_9 = I_9 + 2.$$

The residual two is not an arbitrary adjustment: it is exactly the Euler characteristic of the spherical boundary. This equality is the strongest purely combinatorial selection rule in the extracted definition. It does not require reciprocal fitting, material constants, or the magnetic-moment comparison.

10. Interface Hierarchy and Cell-Center Cycle Rank

10.1 Exact 3:1:1 decomposition

The 1944 internal interfaces can be classified by whether they lie inside a 3B Lucas Group or glue two Lucas Groups together. The outer surface contributes a third face population.

$$N_{\text{intragroup}} = 27 [3(3-1)3^2] = 1458. \quad N_{\text{intergroup}} = 3(3-1)9^2 = 486. \quad N_{\text{external}} = 6(9^2) = 486.$$

$$1458 : 486 : 486 = 3 : 1 : 1.$$

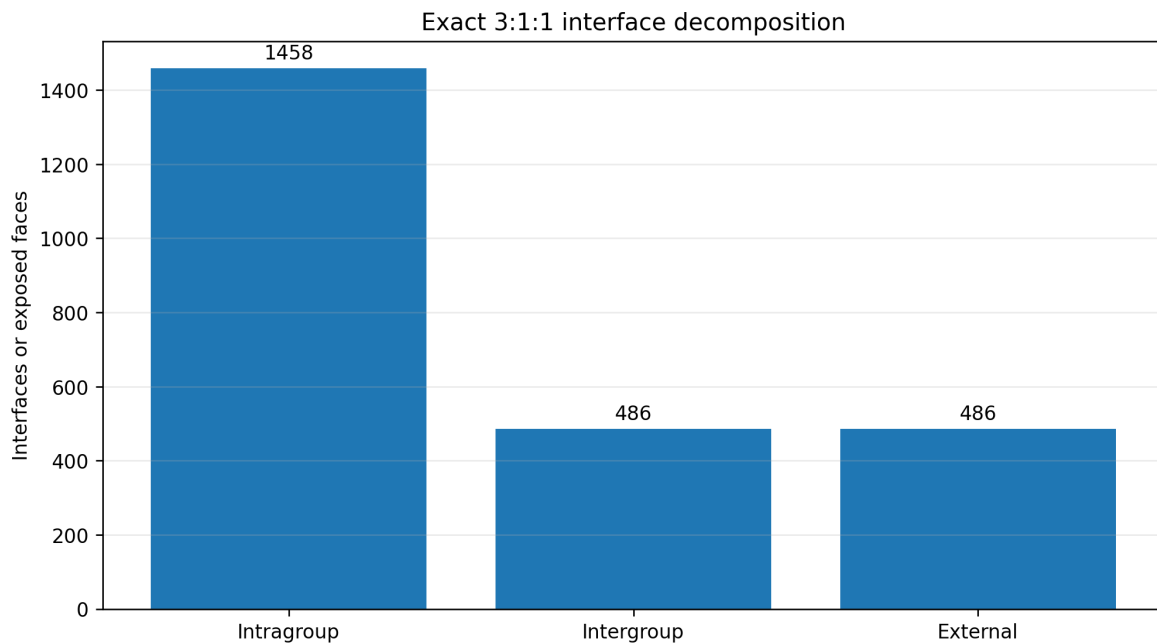


Figure 6. Intragroup interfaces, intergroup gluing interfaces, and exposed faces form the exact ratio 3:1:1.

10.2 Graph cycle rank

Construct a graph whose vertices are the 729 Cell3 volumes and whose edges are the 1944 internal Cell3 interfaces. The block is connected, so its first graph Betti number or cycle rank is

$$\text{beta}_1 = E - V + 1 = 1944 - 729 + 1 = 1216.$$

The value 1216 belongs to this 729-vertex volume-adjacency graph. It is not the cycle rank of the 1000-vertex cubical 1-skeleton, which is a different graph. Preserving this distinction is necessary whenever graph invariants are compared with cellular invariants.

11. Consecutive Cubic Ladders and Finite Derivatives

11.1 Unrefined and refined ladders

At $m=9$, the complete and interior populations organize into consecutive perfect cubes. The unrefined ladder is

$$7^3=343, 8^3=512, 9^3=729, 10^3=1000.$$

These values respectively count strict-interior Cell3 volumes, strict-interior vertices, total Cell3 volumes, and total vertices. The refined ladder is

$$16^3=4096, 17^3=4913, 18^3=5832, 19^3=6859.$$

The 17^3 and 19^3 terms are the strict-interior and complete half-step state counts. The intermediate even counts provide the consecutive cubic context in which these two odd refined lattices sit.

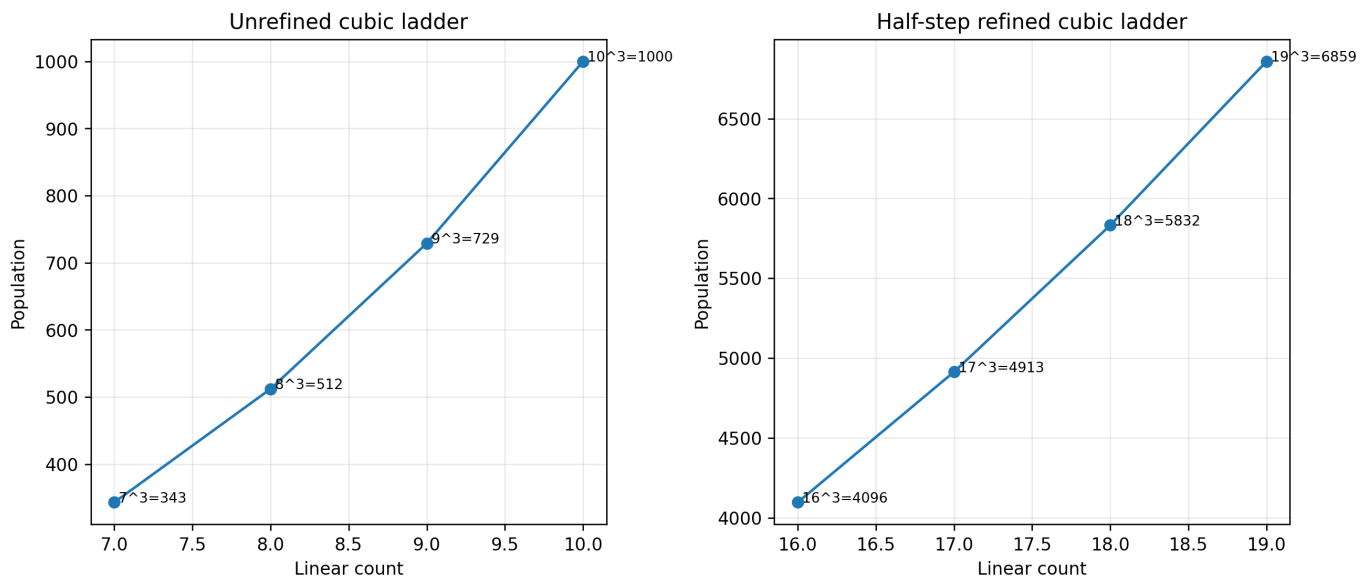


Figure 7. Consecutive cubic ladders at the unrefined and half-step-refined levels.

11.2 Finite geometric derivatives

For the cubic volume law $V(m)=m^3$, the forward first difference, centered second difference, and third difference at the Supergroup scale are

$$\Delta V(9)=V(10)-V(9)=271. \quad V(10)-2V(9)+V(8)=54. \quad \Delta^3 V=6.$$

The hierarchy 729, 271, 54, 6 records the cubic population, its first discrete increment, its centered curvature, and the constant third finite difference. The value 271 reappears as the central refined-sheet population, linking the first volume increment with the largest constant-sum section of the complete cellular-state lattice.

12. Exact Refined Laminar Population Law

12.1 Refined address lattice

$$U_9 = \{-9, -8, \dots, 8, 9\}^3, \quad |U_9|=19^3=6859.$$

For u in U_9 , the laminar index $\lambda=u_1+u_2+u_3$ ranges from -27 to +27. There are therefore 55 possible sheet values. The population $P_9(\lambda)$ counts the number of integer triples in the cube with the specified sum.

12.2 Piecewise closed form

$$P_9(\lambda) = 271 - \lambda^2, \quad |\lambda| \leq 9. \quad P_9(\lambda) = C(29 - |\lambda|, 2), \quad 9 \leq |\lambda| \leq 27.$$

The central region is quadratic because all three coordinates still have room to vary without reaching opposite truncation boundaries. The tails are triangular-number sections because the sum plane cuts a shrinking corner simplex. The formulas agree at $|\lambda|=9$: $271-81=190=C(20,2)$.

$$P_9(0)=271, \quad P_9(\lambda)=P_9(-\lambda), \quad \sum P_9(\lambda)=6859.$$



Figure 8. Complete population of all 55 refined laminar sheets. Dashed boundaries at $\lambda = \pm 9$ separate the quadratic core from the triangular tails.

12.3 Antipodal census

The refined lattice has one inversion-fixed point, $u=0$. Every other state belongs to a two-element antipodal orbit. Hence

$$(6859-1)/2 = 3429 \text{ antipodal pairs.}$$

The 55-sheet count and the central population 271 belong to the refined 19^3 cellular-state lattice. The 729 Cell3 centers occupy only the 25 sums from -12 through +12. This is a level distinction, not a contradiction.

13. Recursive Chebyshev Shell Hierarchy

13.1 Shell definition and population

$$S_r = \{u \text{ in } Z^3 : ||u||_{\infty} = r\}.$$

The shell is the outer boundary of the integer cube $[-r, r]^3$. Its directed population is the full cube minus the cube one unit smaller in every direction:

$$|S_r| = (2r+1)^3 - (2r-1)^3 = 24r^2 + 2.$$

No shell state is fixed by inversion when $r > 0$, so the antipodal population is $12r^2 + 1$.

Radius	Directed shell states	Antipodal pairs	Primitive directed	Primitive pairs	Inherited pairs
1	26	13	26	13	0
3	218	109	192	96	13
9	1946	973	1728	864	109

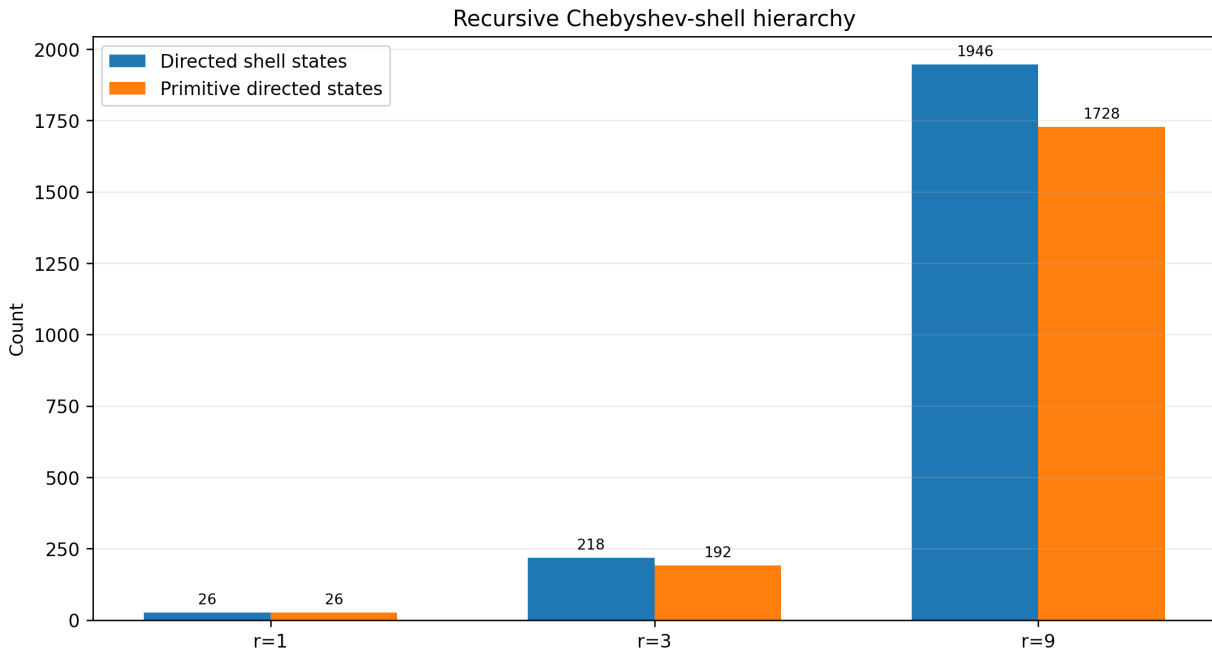


Figure 9. Directed and primitive shell populations at recursive radii 1, 3, and 9.

13.2 Primitive directions and the Jordan totient

A shell address is primitive when the greatest common divisor of the absolute coordinate values is one. For $r > 1$, the primitive directed population is governed by the second Jordan totient $J_2(r)$:

$$J_2(r) = r^2 \prod_{p|r} (1 - 1/p^2), \quad |S_r^{\text{prim}}| = 24 J_2(r).$$

At $r=9$, $J_2(9) = 81(1 - 1/9) = 72$. Therefore the directed primitive population is 1728 and the antipodal primitive population is

$$1728/2 = 864.$$

The remaining 218 directed radius-nine shell states are inherited from the radius-three shell by multiplication by three. The identity $|S_9| = 1946$ is the address-space form of the complete boundary-state census derived earlier.

14. Primitive Reciprocal-Shell Metric

14.1 Quadratic observation law

Every integer direction u receives a reciprocal metric value

$$q(u) = u^T R u.$$

Because R is symmetric, q depends on six independent metric coefficients. Define the observation vector

$$\phi(u) = [u_1^2, u_2^2, u_3^2, 2u_1u_2, 2u_1u_3, 2u_2u_3]^T.$$

$$q(u) = \phi(u)^T [R_{11}, R_{22}, R_{33}, R_{12}, R_{13}, R_{23}]^T.$$

Selecting one representative from each of the 864 primitive antipodal directions produces an 864×6 observation matrix O . Its rank is six. Therefore the shell samples every independent component of a symmetric three-dimensional quadratic metric.

$$\text{rank}(O) = 6, \quad \text{compatibility dimension} = 864 - 6 = 858.$$

The compatibility dimension means that an arbitrary list of 864 numbers cannot be the q -spectrum of one symmetric 3D metric. The list must satisfy 858 independent linear relations. The Supergroup shell is therefore a heavily overdetermined finite directional spectrometer.

14.2 Exact q statistics

Statistic	Value
Count	864
Unique at 1e-10 precision	864
Minimum	130.593523477839227 nm ⁻² at representative (1,4,9)
Maximum	1744.361292059535117 nm ⁻² at representative (9,9,-8)
Mean	757.916780244175357 nm ⁻²
Population standard deviation	325.034198848753931 nm ⁻²
trace(R)	16.842595116537229 nm ⁻²
Mean / trace(R)	45.0
Coordinate square sums	[38880, 38880, 38880]

The primitive shell is coordinate-isotropic in second moments: each coordinate contributes 38880 to the square sum and all cross sums vanish. Consequently the mean q-value is exactly 45 trace(R). The 864 q-values are all distinct at the stated numerical precision after antipodal inversion has already been quotiented out.

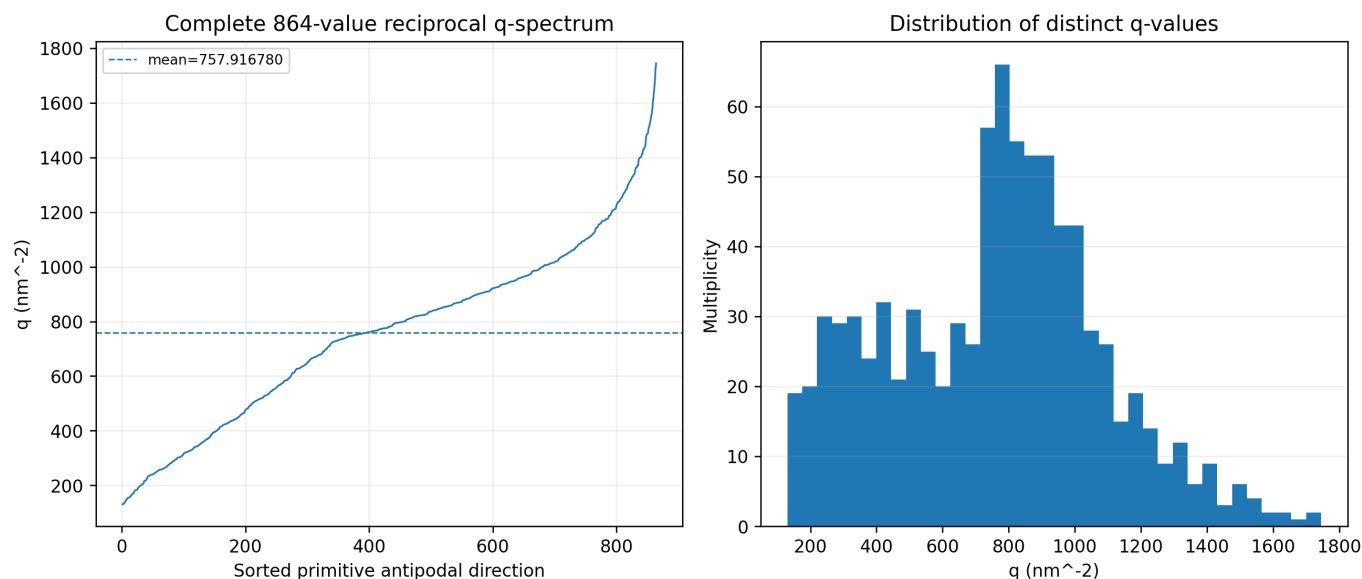


Figure 10. Sorted and histogram views of the complete 864-value primitive antipodal q-spectrum.

15. Metric Observation Sectors

The Gram matrix $O^T O$ separates the six metric coordinates into irreducible sectors under the shell symmetries: one trace scalar, two diagonal-traceless components, and three off-diagonal shear components. The squared singular values are

Sector	Squared singular value	Multiplicity
Trace scalar	5,584,752	1
Diagonal traceless	1,175,976	2
Off-diagonal shear	5,878,368	3

The multiplicities $1+2+3=6$ exhaust the symmetric metric space. The three sectors are not fitted categories; they arise from the exact eigencluster structure of the observation Gram matrix.

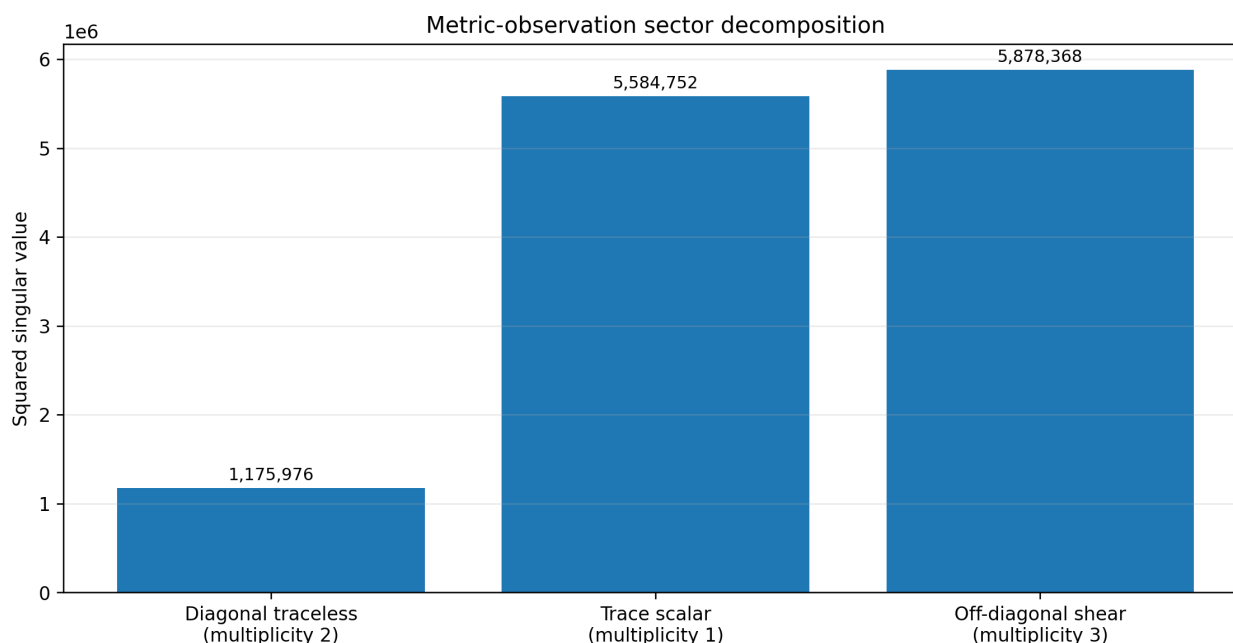


Figure 11. Exact metric-sector squared singular values and multiplicities.

16. Reciprocal Principal System

16.1 Eigenvalues and direct lengths

Diagonalizing R identifies the soft, intermediate, and hard reciprocal directions. If ρ_i is a reciprocal eigenvalue, then the corresponding direct eigenvalue is $\mu_i = 1/\rho_i$ and the associated Cell3 principal length is $\sqrt{\mu_i}$. Multiplying that length by nine gives the Supergroup principal span.

Mode	ρ (nm ⁻²)	μ (nm ²)	Cell3 length (nm)	9B span (nm)	Best radius-9 integer address	Angular error (deg)	Relative eigenvalue error
soft	1.222872851445331	0.817746504731122	0.904293373154488	8.138640358390392	(-1, -6, -9)	2.046273682	0.007084675
intermediate	6.333432714031713	0.157892259245843	0.397356589533687	3.576209305803186	(-1, 9, -6)	2.460791965	-0.000495789
hard	9.286289551060182	0.107685636389168	0.328154896945281	2.953394072507534	(-9, -1, 1)	2.706425197	-0.001305245

condition number $\kappa(R) = \rho_{\text{hard}}/\rho_{\text{soft}} = 7.593830822300602$.

The stored eigenvectors are expressed in primitive-coordinate space. A primitive-coordinate direction is not itself an ICRS unit vector; conversion to physical or reciprocal Cartesian direction requires the corresponding basis transformation. The best integer eigenaddresses are finite shell approximations, not exact eigenvectors.

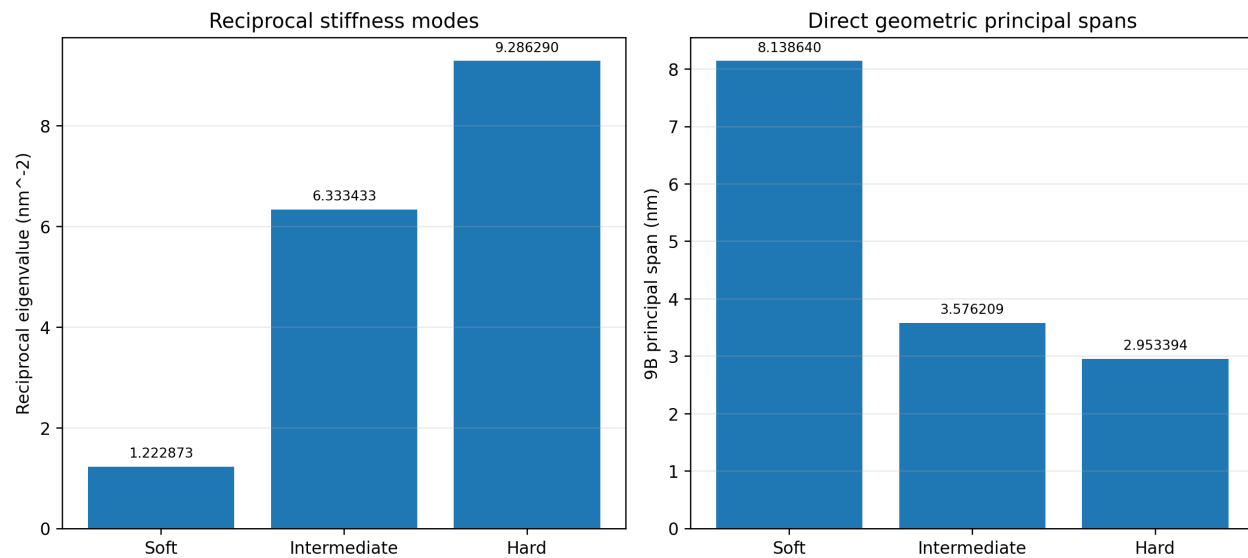


Figure 12. Reciprocal eigenvalues and corresponding direct 9B principal spans.

17. Super-Emily Spectral Hierarchy

17.1 Reconstructed nearest-neighbor operator

The reported spectral extrema exactly identify the standard open cubic nearest-neighbor adjacency operator on $P_n \times P_n \times P_n$. For a scalar state $f(i,j,k)$, the operator sums the existing six axial neighbors:

$$(A_n f)_{ijk} = f_{i-1,j,k} + f_{i+1,j,k} + f_{i,j-1,k} + f_{i,j+1,k} + f_{i,j,k-1} + f_{i,j,k+1}.$$

Terms outside the finite cube are omitted. The separable eigenvalues are

$$\lambda_{pqr} = 2[\cos(p\pi/(n+1)) + \cos(q\pi/(n+1)) + \cos(r\pi/(n+1))], \quad 1 \leq p, q, r \leq n.$$

$$\lambda_{\max} = 6 \cos(\pi/(n+1)), \quad \lambda_{\min} = -\lambda_{\max}.$$

17.2 Hierarchy-relevant spectra

Role	n	States n^3	Sheets $3n-2$	Nullity	Rank	Minimum eigenvalue	Maximum eigenvalue
Cell3 half-step graph	3	27	7	7	20	-4.242640687119285	4.242640687119286
Lucas Group half-step graph	7	343	19	19	324	-5.543277195067720	5.543277195067720
Supergroup center graph	9	729	25	25	704	-5.706339097770921	5.706339097770921
Supergroup half-step graph	19	6859	55	55	6804	-5.926130043570826	5.926130043570827

At $n=3, 7, 9$, and 19 , the reported adjacency nullity equals the number $3n-2$ of constant-sum laminar sheets. For the complete refined Supergroup lattice, the result is

$$\dim \ker(A_{19}) = 55, \quad \text{rank}(A_{19}) = 6804.$$

The equality between sheet count and nullity is an exact reported property of these four hierarchy sizes. The source data specify the spectra but do not name the operator formula; the open-cube adjacency identification is reconstructed because its complete eigenvalue formula reproduces all stored extrema exactly.

Open cubic nearest-neighbor adjacency spectra at hierarchy-relevant sizes

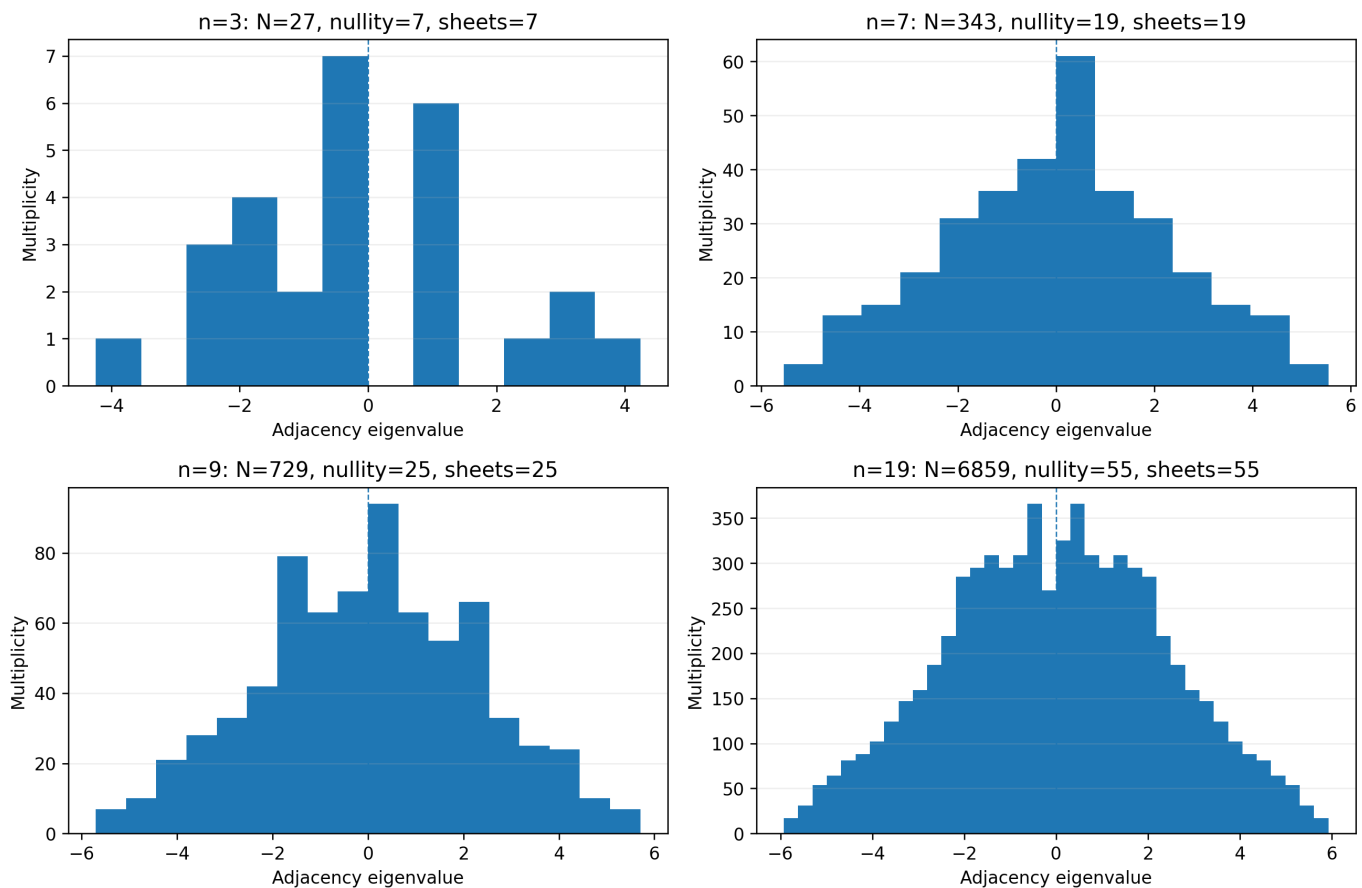


Figure 13. Adjacency spectra computed from the reconstructed open-cube eigenvalue formula at $n=3, 7, 9$, and 19 .

18. The Geometric Integer 864 and the Physics Candidate

The primitive radius-nine shell contains exactly 864 antipodally distinct directions. Its prime factorization is

$$864 = 2^5 \times 3^3.$$

The source compares this integer with the electron-to-shielded-helion magnetic-moment ratio 864.058255. The numerical residual is

$$(864 - 864.058255) / 864.058255 = -6.742022272565595e-5, \text{ absolute residual} = 67.42022272565595 \text{ ppm}.$$

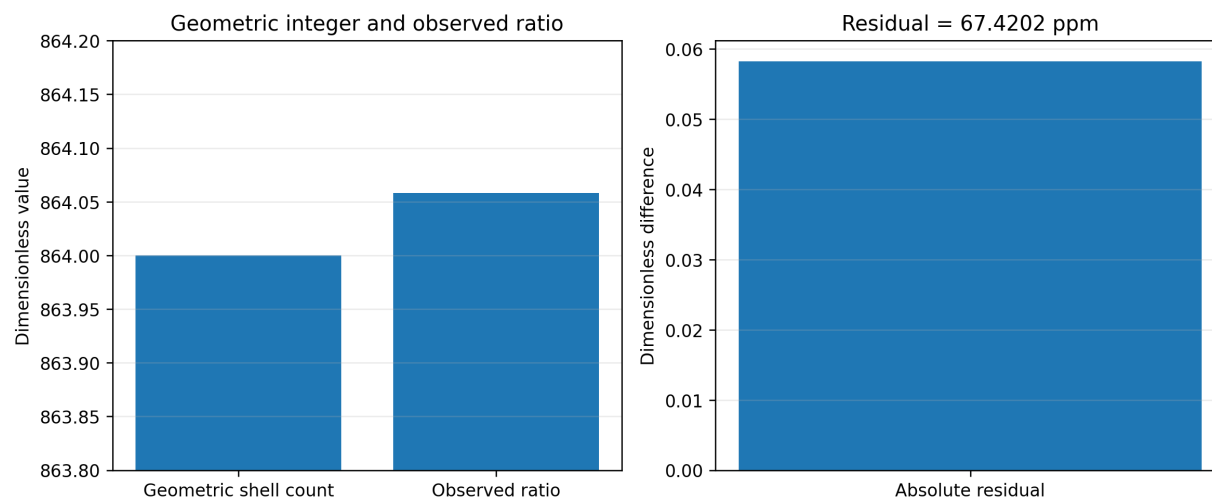


Figure 14. The shell integer 864 compared with the reported observed ratio. The expanded vertical scale makes the small difference visible.

The exact result is the geometric origin of 864 as the number of primitive antipodal directions on the radius-nine outer shell. The source does not contain a coupling operator that maps this shell count to the measured magnetic-moment ratio. The comparison is therefore a candidate identification and a target for future derivation, not a completed consequence of the presently defined geometry.

19. Canonical Axiom System

The complete object can be generated from the following compact axiom system. Each axiom is operational: it states a finite set, a map, or an algebraic relation sufficient for independent reconstruction.

Axiom	Statement
A1 - Ordered metric basis	B is the authoritative 3 x 3 direct basis with columns a,b,c; $M=B^T B$ and $R=M^{-1}$.
A2 - Local half-step embedding	Cell3 scaffold addresses are u in $\{-1,0,1\}^3$ and $r(u)=B u/2$.
A3 - Cellular class	The number of zero address coordinates determines corner, edge-center, face-center, or body-center class.
A4 - Laminar grade	$\lambda(u)=u_1+u_2+u_3$.
A5 - Polarity	$\chi(u)=(-1)^\lambda(u)$.
A6 - Antipodal involution	$A(u)=-u$; $r(-u)=-r(u)$; $\lambda(-u)=-\lambda(u)$; $\chi(-u)=\chi(u)$.
A7 - Boundary planes	$e_i^T B^{-1} r = \pm 1/2$ for $i=a,b,c$.
A8 - Recursive balanced ternary	Every Supergroup center address has unique form $x=3G+c$ with G,c in $\{-1,0,1\}^3$.
A9 - Physical center embedding	$r_{\text{center}}(x)=B x$.
A10 - Phase quotient	Periodic address classes form Z_9^3 and satisfy $0 \rightarrow Z_3^3 \rightarrow Z_9^3 \rightarrow Z_3^3 \rightarrow 0$, nonsplit.
A11 - Cellular complex	The m-block f-polynomial is $F_m(t)=((m+1)+mt)^3$; the Lucas Supergroup uses $m=9$.
A12 - Refined state lattice	$U_9=\{-9,\dots,9\}^3$ represents the barycenters of all cells and contains 19^3 states.
A13 - Shell and primitive quotient	$S_r=\{u: \ u\ _\infty=r\}$; primitive addresses have $\gcd(u_1 , u_2 , u_3)=1$; antipodal directions identify u with $-u$.
A14 - Reciprocal observation	$q(u)=u^T R u$.
A15 - Nearest-neighbor spectrum	The reconstructed Emily operator is open cubic nearest-neighbor adjacency on hierarchy-relevant n .

20. Theorem and Result Ledger

Result	Exact statement
Primitive census	27 nodes = 8 corners + 12 edge centers + 6 face centers + 1 body center.
Primitive sheets	$\lambda = -3, \dots, 3$ gives seven Cell3 sheets.
Lucas Group	3B block = 27 Cell3 volumes.
Lucas Supergroup	9B block = 27 Lucas Groups = 729 Cell3 volumes.
Complete f-vector	(1000,2700,2430,729), Euler characteristic 1.
Interior f-vector	(512,1728,1944,729), compact-support Euler value -1.
Boundary f-vector	(488,972,486), total 1946, Euler characteristic 2.
Unique closure scale	$(2m+1)^3 - (2m-1)^3 = 3m^2(m-1) + 2$ has unique positive solution $m=9$.
Closure identity	$1946 = 1944 + 2$.
Interface decomposition	$1458:486:486 = 3:1:1$.
Cell-center graph cycles	$1944 - 729 + 1 = 1216$.
Cubic ladders	343,512,729,1000 and 4096,4913,5832,6859.
Finite derivatives	$V_9 = 729$, forward first=271, centered second=54, third difference=6.
Refined sheets	55 sheets; $P_9(0) = 271$; exact quadratic-core and triangular-tail law.
Refined antipodal pairs	$(6859 - 1)/2 = 3429$.
Radius-nine shell	1946 directed states, 973 antipodal pairs.
Primitive shell	1728 directed primitive states, 864 primitive antipodal directions.
Observation rank	$\text{rank}(O) = 6$ and compatibility dimension 858.
Mean q identity	$\text{mean}(q) = 45$ $\text{trace}(R) = 757.9167802441754 \text{ nm}^{-2}$.
Reciprocal anisotropy	condition number 7.593830822300602.
Refined Emily system	$n=19$, 6859 states, 55 sheets, nullity 55, rank 6804.

21. Interpretive Precision and Scope

Five distinctions are required to preserve the definition without category errors.

- **6859 is a complete barycentric-state count.** It combines vertices, edge barycenters, face barycenters, and volume centers; it is not the ordinary vertex count.
- **Fifty-five sheets belong to the refined $19 \times 19 \times 19$ state lattice.** The 729 Cell3 centers occupy 25 laminar sums.
- **The cycle rank 1216 belongs to the Cell3 adjacency graph.** It is not the cycle rank of the cubical one-skeleton.
- **Z_9^3 is a periodic phase group.** The bounded representative cube $\{-4, \dots, 4\}^3$ is not closed under ordinary addition.
- **The present source fixes geometry, arithmetic, topology, shells, reciprocal metrics, and spectra.** It does not independently fix a weighted gluing law, an edge-plaquette-volume action, holonomy, a dynamical energy functional, or a physical coupling operator.

These distinctions do not weaken the object. They establish exactly which future developments are constrained by the extracted definition. Any subsequent dynamical theory must reduce to the basis, counts, involutions, shell census, and metric spectra defined here.

22. Compact Canonical Definition

L_9 is the $9 \times 9 \times 9$ oblique cubical complex generated by the authoritative basis B . Its Cell3 centers are Bx with $x = 3G + c$ and G, c in $\{-1, 0, 1\}^3$. It contains 729 Cell3 volumes and 27 Lucas Groups. Its cubical f-polynomial is $(10+9t)^3$ and its f-vector is (1000,2700,2430,729). Its complete barycentric lattice is $U_9 = \{-9, \dots, 9\}^3$ with 6859 states, laminar grade $\lambda(u) = u_1 + u_2 + u_3$, polarity $\chi(u) = (-1)^\lambda$, and antipodal involution $u \rightarrow -u$. Its boundary contains 1946 barycentric states and its internal Cell3 adjacency has 1944 edges and cycle rank 1216. Its primitive radius-nine shell has 864 antipodal directions. Its reciprocal directional law is $q(u) = u^T R u$ with $R = (B^T B)^{-1}$. The unique positive interface-boundary closure scale is $m=9$, expressed by $1946 = 1944 + 2$.

23. Educational Consequences and Research Pathways

The extracted definition changes the role of the Lucas hierarchy from a qualitative aggregation picture into a reproducible finite mathematics. Students can now move from one matrix B to all direct dimensions, every local node, each boundary plane, the complete 729-volume arrangement, all cellular counts, the exact boundary topology, the laminar distribution, the primitive shell, and the reciprocal metric spectrum without introducing independent geometric parameters.

Several research pathways follow directly. A dynamical extension must specify how states on vertices, edges, faces, or volumes evolve while respecting the antipodal and parity structure. A field extension must define a controlled continuum limit from recursive finite blocks while preserving the exact finite closure identity. A physical-coupling extension must map shell or graph observables to measured quantities with explicit dimensions and testable operators. A spectral extension must explain why the reported hierarchy sizes have nullity equal to sheet count and whether that equality persists under metric weighting. Each pathway begins from a fixed object rather than from an unconstrained analogy.

The central methodological advance is therefore deterministic closure. Direct geometry, reciprocal geometry, integer address arithmetic, topology, and finite spectra are no longer separate descriptive layers. They are different projections of the same 9B construction.

Appendix A. Complete 27-Node Cell3 Ledger

Node numbers correspond to Figures 2 and 3. Coordinates are in ICRS nanometres and are generated by $r(u)=Bu/2$.

ID	Address	lambda	Polarity	Class	x (nm)	y (nm)	z (nm)
01	(-1, -1, -1)	-3	-1	corner	-0.176351350136495	0.118682429905037	0.637311498325253
02	(-1, -1, 0)	-2	1	edge center	-0.181416008949859	-0.076211421278093	0.292444294141619
03	(-1, -1, 1)	-1	-1	corner	-0.186480667763224	-0.271105272461223	-0.052422910042015
04	(-1, 0, -1)	-2	1	edge center	-0.158685962214581	0.222342355125133	0.363610955630999
05	(-1, 0, 0)	-1	-1	face center	-0.163750621027945	0.027448503942003	0.018743751447365
06	(-1, 0, 1)	0	1	edge center	-0.168815279841310	-0.167445347241127	-0.326123452736269
07	(-1, 1, -1)	-1	-1	corner	-0.141020574292667	0.326002280345229	0.089910412936745
08	(-1, 1, 0)	0	1	edge center	-0.146085233106031	0.131108429162099	-0.254956791246889
09	(-1, 1, 1)	1	-1	corner	-0.151149891919396	-0.063785422021031	-0.599823995430523
10	(0, -1, -1)	-2	1	edge center	-0.012600729108550	0.091233925963034	0.618567746877888
11	(0, -1, 0)	-1	-1	face center	-0.017665387921914	-0.103659925220096	0.273700542694254
12	(0, -1, 1)	0	1	edge center	-0.022730046735279	-0.298553776403226	-0.071166661489380
13	(0, 0, -1)	-1	-1	face center	0.005064658813364	0.194893851183130	0.344867204183634
14	(0, 0, 0)	0	1	body center	0.000000000000000	0.000000000000000	0.000000000000000
15	(0, 0, 1)	1	-1	face center	-0.005064658813364	-0.194893851183130	-0.344867204183634
16	(0, 1, -1)	0	1	edge center	0.022730046735279	0.298553776403226	0.071166661489380
17	(0, 1, 0)	1	-1	face center	0.017665387921914	0.103659925220096	-0.273700542694254
18	(0, 1, 1)	2	1	edge center	0.012600729108550	-0.091233925963034	-0.618567746877888
19	(1, -1, -1)	-1	-1	corner	0.151149891919396	0.063785422021031	0.599823995430523
20	(1, -1, 0)	0	1	edge center	0.146085233106031	-0.131108429162099	0.254956791246889
21	(1, -1, 1)	1	-1	corner	0.141020574292667	-0.326002280345229	-0.089910412936745
22	(1, 0, -1)	0	1	edge center	0.168815279841310	0.167445347241127	0.326123452736269
23	(1, 0, 0)	1	-1	face center	0.163750621027945	-0.027448503942003	-0.018743751447365
24	(1, 0, 1)	2	1	edge center	0.158685962214581	-0.222342355125133	-0.363610955630999
25	(1, 1, -1)	1	-1	corner	0.186480667763224	0.271105272461223	0.052422910042015
26	(1, 1, 0)	2	1	edge center	0.181416008949859	0.076211421278093	-0.292444294141619
27	(1, 1, 1)	3	-1	corner	0.176351350136495	-0.118682429905037	-0.637311498325253

Appendix B. Complete Six-Plane Ledger

Plane	Family	Primitive coordinate	Normal (ICRS)	Area nm^2	Spacing nm	Center distance nm	Nine node addresses
a-	bc	-0.5	(-0.995965190, 0.083602163, -0.032619291)	0.357809539128562	0.329546529695447	0.164773264847723	(-1, -1, -1); (-1, -1, 0); (-1, -1, 1); (-1, 0, -1); (-1, 0, 0); (-1, 0, 1); (-1, 1, -1); (-1, 1, 0); (-1, 1, 1)
a+	bc	0.5	(-0.995965190, 0.083602163, -0.032619291)	0.357809539128562	0.329546529695447	0.164773264847723	(1, -1, -1); (1, -1, 0); (1, -1, 1); (1, 0, -1); (1, 0, 0); (1, 0, 1); (1, 1, -1); (1, 1, 0); (1, 1, 1)
b-	ac	-0.5	(-0.089052594, -0.866576769, 0.491033948)	0.261106235790205	0.451597379721285	0.225798689860642	(-1, -1, -1); (-1, -1, 0); (-1, -1, 1); (0, -1, -1); (0, -1, 0); (0, -1, 1); (1, -1, -1); (1, -1, 0); (1, -1, 1)
b+	ac	0.5	(-0.089052594, -0.866576769, 0.491033948)	0.261106235790205	0.451597379721285	0.225798689860642	(-1, 1, -1); (-1, 1, 0); (-1, 1, 1); (0, 1, -1); (0, 1, 0); (0, 1, 1); (1, 1, -1); (1, 1, 0); (1, 1, 1)
c-	ab	-0.5	(0.194092201, 0.913177169, 0.358379235)	0.194869165460484	0.605097741518555	0.302548870759278	(-1, -1, -1); (-1, 0, -1); (-1, 1, -1); (0, -1, -1); (0, 0, -1); (0, 1, -1); (1, -1, -1); (1, 0, -1); (1, 1, -1)
c+	ab	0.5	(0.194092201, 0.913177169, 0.358379235)	0.194869165460484	0.605097741518555	0.302548870759278	(-1, -1, 1); (-1, 0, 1); (-1, 1, 1); (0, -1, 1); (0, 0, 1); (0, 1, 1); (1, -1, 1); (1, 0, 1); (1, 1, 1)

Appendix C. Complete 55-Sheet Population Ledger

lambda	P9(lambda)	Formula sector
-27	1	triangular tail: $C(29- \lambda , 2)$
-26	3	triangular tail: $C(29- \lambda , 2)$
-25	6	triangular tail: $C(29- \lambda , 2)$
-24	10	triangular tail: $C(29- \lambda , 2)$
-23	15	triangular tail: $C(29- \lambda , 2)$
-22	21	triangular tail: $C(29- \lambda , 2)$
-21	28	triangular tail: $C(29- \lambda , 2)$
-20	36	triangular tail: $C(29- \lambda , 2)$
-19	45	triangular tail: $C(29- \lambda , 2)$
-18	55	triangular tail: $C(29- \lambda , 2)$
-17	66	triangular tail: $C(29- \lambda , 2)$
-16	78	triangular tail: $C(29- \lambda , 2)$
-15	91	triangular tail: $C(29- \lambda , 2)$
-14	105	triangular tail: $C(29- \lambda , 2)$
-13	120	triangular tail: $C(29- \lambda , 2)$
-12	136	triangular tail: $C(29- \lambda , 2)$
-11	153	triangular tail: $C(29- \lambda , 2)$
-10	171	triangular tail: $C(29- \lambda , 2)$
-9	190	quadratic core: $271-\lambda^2$
-8	207	quadratic core: $271-\lambda^2$
-7	222	quadratic core: $271-\lambda^2$
-6	235	quadratic core: $271-\lambda^2$
-5	246	quadratic core: $271-\lambda^2$
-4	255	quadratic core: $271-\lambda^2$
-3	262	quadratic core: $271-\lambda^2$
-2	267	quadratic core: $271-\lambda^2$
-1	270	quadratic core: $271-\lambda^2$
0	271	quadratic core: $271-\lambda^2$
1	270	quadratic core: $271-\lambda^2$
2	267	quadratic core: $271-\lambda^2$
3	262	quadratic core: $271-\lambda^2$
4	255	quadratic core: $271-\lambda^2$
5	246	quadratic core: $271-\lambda^2$
6	235	quadratic core: $271-\lambda^2$
7	222	quadratic core: $271-\lambda^2$
8	207	quadratic core: $271-\lambda^2$
9	190	quadratic core: $271-\lambda^2$
10	171	triangular tail: $C(29- \lambda , 2)$
11	153	triangular tail: $C(29- \lambda , 2)$
12	136	triangular tail: $C(29- \lambda , 2)$
13	120	triangular tail: $C(29- \lambda , 2)$
14	105	triangular tail: $C(29- \lambda , 2)$
15	91	triangular tail: $C(29- \lambda , 2)$
16	78	triangular tail: $C(29- \lambda , 2)$
17	66	triangular tail: $C(29- \lambda , 2)$
18	55	triangular tail: $C(29- \lambda , 2)$
19	45	triangular tail: $C(29- \lambda , 2)$

lambda	P9(lambda)	Formula sector
20	36	triangular tail: $C(29- \lambda , 2)$
21	28	triangular tail: $C(29- \lambda , 2)$
22	21	triangular tail: $C(29- \lambda , 2)$
23	15	triangular tail: $C(29- \lambda , 2)$
24	10	triangular tail: $C(29- \lambda , 2)$
25	6	triangular tail: $C(29- \lambda , 2)$
26	3	triangular tail: $C(29- \lambda , 2)$
27	1	triangular tail: $C(29- \lambda , 2)$

Appendix D. Principal Reciprocal Modes and Integer Approximants

Mode	Primitive-coordinate eigenvector	Best integer address	Rayleigh quotient nm^{-2}	Angular error deg	Relative error
soft	(-0.068132680913, -0.531194531322, -0.844505954796)	(-1, -6, -9)	1.231536507592557	2.046273682327	0.007084674533
intermediate	(-0.117594007582, 0.844848183705, -0.521922595670)	(-1, 9, -6)	6.330292669201262	2.460791965033	-0.000495788772
hard	(-0.990721750631, -0.063748853979, 0.120027065464)	(-9, -1, 1)	9.274168671425066	2.706425196568	-0.001305244637

Appendix E. Formula Ledger

Domain	Formula
Local embedding	$r(u) = B \cdot u/2$
Direct metric	$M = B^T B$
Reciprocal metric	$R = M^{-1}$
Laminar grade	$\lambda(u) = u_1 + u_2 + u_3$
Polarity	$\chi(u) = (-1)^{\lambda(u)}$
Balanced ternary	$x = 3G + c$
Cell-center embedding	$r_{\text{center}}(x) = Bx$
Scale laws	$B_s = sB$; $M_s = s^2M$; $R_s = s^{-2}R$; $A_s = s^2A$; $V_s = s^3V$
Cubical generating polynomial	$F_m(t) = ((m+1) + mt)^3$
Complete state count	$F_m(1) = (2m+1)^3$
Interior polynomial	$F_m^{\text{int}}(t) = ((m-1) + mt)^3$
Boundary state count	$B_m = (2m+1)^3 - (2m-1)^3 = 24m^2 + 2$
Internal interfaces	$I_m = 3m^2(m-1)$
Closure equation	$B_m = I_{m+2} \rightarrow m=9$
Cell-center cycle rank	$\beta_1 = I_9 - 9^3 + 1 = 1216$
Sheet core	$P_9(\lambda) = 271 - \lambda^2$ for $ \lambda \leq 9$
Sheet tail	$P_9(\lambda) = C(29 - \lambda , 2)$ for $9 < \lambda \leq 27$
Chebyshev shell	$ S_r = 24r^2 + 2$
Jordan primitive shell	$ S_r^{\text{prim}} = 24J_2(r)$
Reciprocal observation	$q(u) = u^T R u$
Observation vector	$\phi(u) = [u_1^2, u_2^2, u_3^2, 2u_1u_2, 2u_1u_3, 2u_2u_3]^T$
Emily eigenvalues	$\lambda_{pqr} = 2[\cos(p \pi/(n+1)) + \cos(q \pi/(n+1)) + \cos(r \pi/(n+1))]$

Appendix F. Source Record and Reproducibility

Source	Role	SHA-256
K-Field_Mass-Bridge_is_now_a_locked_projectiv e_ratio_system_1785176592(1).json	Locked Cell3 and Mass-Bridge source	a5ba3ddf21a20e4528c19609a3061fceca60f53f76224f101 c9779c24809d7ea
K-Field_mass_bridge_drives_volume_filling_hiera rchy_1785204375.json	Prior volume-filling hierarchy and physics comparison source	811c98f49447534c27d14fd77c779d756e4acc73bf44bfb9 3ae7bebf81e79b2
K-Field_Lucas_Supergroup_Analysis_178520801 4(1).json	Authoritative loaded analysis used for all numerical tables and figures	37efaa688f9cb496ae16c76af6f7889bd78e10e4e27c15cee 9b3e2cd3331f6b5

All figures in this edition are regenerated from the numerical records in the authoritative JSON. The base-cell and boundary-plane graphics use the stored ICRS coordinates and plane incidences. The Supergroup graphic is generated from the balanced-ternary center addresses and the 9B boundary. Population charts use the exact stored counts. The Emily spectral plots use the uniquely reconstructed open-cube nearest-neighbor eigenvalue formula that reproduces the stored extrema.

The machine-readable companion preserves the complete authoritative numerical data, the canonical axiom system, the formula and theorem ledgers, generated figure manifest, and document metadata. It is intended to be the computational source for independent reproduction of this paper.